Homework 5: Liquids & pH

Paper track. Pairs with Home Lab 5. Do this on your own.

Part A: Identify the liquid (2 pts each)

Give the **name and formula**:

1. Clear, sharp sour smell, pH 2.5, leaves no residue when it dries.
2. Cloudy, yellowish, citrus smell, pH 2.
3. Clear, strong pungent smell that stings your nose, pH 11.
4. Clear, faint smell, pH 5, foams when a drop touches raw potato.
5. Clear, medicinal smell, evaporates in seconds off your hand, flammable.
6. Clear, no smell, pH 7, dries to cubic crystals.

Part B: pH (2 pts each)

1. What does pH stand for, and what does a **low** pH mean a solution has more of?
2. The pH scale is logarithmic. How many times more hydrogen ions does pH 2 have than pH 5?
3. Put in order from most acidic to most basic: household ammonia, vinegar, pure water, hydrogen peroxide.
4. Complete the neutralization: `HCl + NaOH → ____ + ____`.

Part C: Cabbage indicator (1 pt each)

Your red-cabbage indicator turns these colors. Acid, neutral, or base? Estimate the pH.

1. Bright pink
2. Purple
3. Green

Part D: Scenario (4 pts)

A note at a crime scene is stained by a spilled clear liquid. The liquid smells sharp and sour, reads pH 2.5 on the paper, is not cloudy, and evaporates leaving no residue. Two suspects are relevant: one pickles vegetables (uses vinegar), one cleans glass (uses ammonia). Which suspect does the liquid implicate, and why is the other cleared?

Answer Key

Part A: 1. Vinegar — CH_3COOH ($C_2H_4O_2$). 2. Lemon juice — ~$C_6H_8O_7$. 3. Ammonia — NH_3. 4. Hydrogen peroxide — H_2O_2. 5. Isopropyl alcohol — C_3H_8O. 6. Salt water — NaCl (in water).

Part B:

1. "Potential of hydrogen"; more **hydrogen ions (H^+)**.
2. 1000 times (3 units × 10 each).
3. vinegar (2.5) → hydrogen peroxide (~5) → pure water (7) → ammonia (11).
4. NaCl + H_2O.

Part C: 1. Acid, ~pH 3–4. 2. Neutral, ~pH 7. 3. Base, ~pH 11.

Part D: The liquid is **vinegar** (clear, sour, pH 2.5, no residue — ammonia would be pH 11 and smell different; lemon juice would be cloudy). It implicates the **pickling** suspect. The glass-cleaning suspect is cleared: their liquid (ammonia) is a base at pH 11, the opposite of what was found.